B.TECH CAPSTONE DEFENSE · ACADEMIC YEAR 2026–2027

TrustChain-AI

A Decentralized Trust Intelligence Framework for Secure Autonomous AI Agent Collaboration

Department of Computer Engineering · Mukesh Patel School of Technology Management & Engineering (MPSTME)
SVKM's NMIMS (Deemed-to-be University), Mumbai

01 · IDENTITY LAYER

W3C did:key & Ed25519

Decentralized cryptographic public-key identity generation without central CA dependency. W3C Verifiable Credentials with URDNA2015 LD-Proofs.

02 · TRUST INTELLIGENCE

Continuous Temporal EMA

Multi-factor runtime telemetry (Success: 60%, Compliance: 30%, Latency: 10%). Mathematical Exponential Moving Average with 4 discrete states.

03 · ZERO-TRUST CONTROL

Adaptive Policy Gateway

Operation-level privilege isolation (Standard vs Privileged). Explainable multi-factor candidate matching and modern React 19 SOC portal.

PROJECT PRESENTERS & MENTORSHIP

Submitted by: Candidate Presenters (B.Tech Computer Engineering) | Mentor & Guide: Project Mentorship Panel

Mukesh Patel School of Technology Management & Engineering (MPSTME), Mumbai | Final Capstone Defense 2026–2027

The Point-in-Time Security Failure in Multi-Agent Systems

Three Critical Failure Modes of Static Authentication

1. The Binary Point-of-Entry Fallacy

Traditional access control treats authentication as a one-time gate: an agent presents a valid credential or API key at t=0 and receives persistent operational access. However, validity at connection time provides zero guarantee of runtime execution safety.

2. Post-Authentication Behavioral Drift

Autonomous agents are prone to model hallucination, API timeout degradation, configuration tampering, and prompt injection subversion. Once past the initial authentication barrier, a degraded agent continues executing privileged workloads undetected.

3. Cascading Compromise in Multi-Agent Workflows

In sequential and collaborative pipelines (Agent A -> Agent B -> Agent C), output from an unmonitored degraded agent is consumed as trusted input by subsequent nodes, compounding errors and undermining systemic integrity.

Security Paradigm Shift: Static vs. Continuous Zero-Trust

TRADITIONAL PARADIGM: BLIND RUNTIME TRUST

```
[Static API Key] -> [Access Granted (t=0)] -> [Unchecked Execution] -> [Cascading Breach]
```

- Binary access: Agent is either 100% trusted or completely blocked.
- No runtime behavioral monitoring or telemetry feedback loop.
- Privileges persist even during catastrophic performance degradation.

TRUSTCHAIN-AI PARADIGM: CONTINUOUS ZERO-TRUST

```
[W3C did:key] -> [Sensor Telemetry] -> [EMA Trust Scoring] -> [Adaptive Gating] -> [Audit]
```

- Multi-factor evaluation: Success rate (S), compliance (C), latency (L).
- Temporal smoothing prevents single-event volatility: $T(t) = 0.7*T(t-1) + 0.3*P(t)$.
- Automatic dynamic isolation: Quarantines degraded agents ($T < 0.50$).

Concrete Vulnerabilities & Threat Vectors in Autonomous Agent Networks

1. RUNTIME DRIFT & HALLUCINATION

Stochastic Model Degradation: Output schemas mutate unexpectedly.

Prompt Injection Subversion: Malicious prompts leak data.

TrustChain Defense: Compliance factor C_i(t) drops to 0.0 upon schema breaches, triggering rapid quarantine.

2. SLA LATENCY & TIMEOUT BREACHES

Network Latency Spikes: Microservice stalls cause pipeline timeouts.

Silent Hangs: Failing agents continue accepting tasks without delivery.

TrustChain Defense: Latency factor L_i(t) enforces normalized SLA penalties against a 2000ms threshold ceiling.

3. CASCADING PIPELINE POISONING

Blast Radius Amplification: Node 1 failure propagates to Node 2.

Byzantine Deception: Degraded agent claims success with bad payload.

TrustChain Defense: Sequential pipeline re-checks live trust at every node and halts immediately if an agent degrades.

4. OVER-PRIVILEGED AGENT DELEGATION

Unchecked Delegation: Standard agents requesting state-altering tools.

Identity Replay: Stolen keys used to execute administrative pipelines.

TrustChain Defense: Cryptographic Ed25519 signatures combined with ABAC: privileged actions strictly blocked in Suspicious/Quarantine.

Theoretical Foundations & Research Synthesis Across Four Domains

DOMAIN 1: SELF-SOVEREIGN IDENTITY

W3C DID Core 1.0 & Verifiable Credentials (2021-2022)

- Strengths: Decentralized public-key identifiers (did:key) without central CA dependency.
- Limitation: Credentials are static point-in-time attestations; zero visibility into runtime behavioral drift.
- Relevance: TrustChain-AI adopts W3C did:key and URDNA2015 JSON-LD credentials as its identity baseline.

DOMAIN 2: DISTRIBUTED REPUTATION SYSTEMS

EigenTrust (Kamvar 2003) & Beta Reputation (Jøsang 2002)

- Strengths: Peer ratings aggregation and historical reliability evaluation in distributed networks.
- Limitation: Classic models ignore SLA execution latency, security compliance, and dynamic API access gating.
- Relevance: TrustChain-AI integrates peer ratings into multi-factor matching, but prioritizes recent telemetry.

DOMAIN 3: ZERO TRUST ARCHITECTURES

NIST SP 800-207 Zero Trust Architecture (2020)

- Strengths: 'Never trust, always verify' paradigm for network perimeters and continuous validation.
- Limitation: Formulated for human enterprise users and network devices; lacks autonomous agent behavioral math.
- Relevance: TrustChain-AI extends Zero Trust to autonomous AI agents via continuous mathematical trust scoring.

DOMAIN 4: MULTI-AGENT DIRECTORIES

FIPA Agent Management Specification (SC00023J)

- Strengths: Standardized Directory Facilitator (DF) yellow-pages agent capability lookups.
- Limitation: Static capability matching without runtime trust scoring, health verification, or explainability.
- Relevance: TrustChain-AI replaces static directories with dynamic discovery and explainable candidate matching.

THE RESEARCH GAP: No prior system unifies W3C decentralized identity, continuous temporal EMA trust evaluation, quad-dimensional audit logging, and adaptive operation-level access control into a closed loop.

Formal Engineering Objectives & Scope Governance

Six Core Technical Objectives (Implemented & Verified)

✓ Obj 1 · Cryptographic Identity Genesis
Implement W3C did:key decentralized identifiers using Ed25519 elliptic-curve keypairs with multicodec encoding (0xED01).

✓ Obj 2 · Verifiable Credential Operations
Formulate W3C JSON-LD Verifiable Credentials with URDNA2015 canonicalization and Ed25519Signature2020 proof verification.

✓ Obj 3 · Multi-Factor Telemetry Ingestion
Capture runtime sensor events evaluating task success rate S_i(t), policy compliance C_i(t), and SLA latency penalty L_i(t).

✓ Obj 4 · Temporal EMA Trust Engine
Develop mathematical Exponential Moving Average smoothing T_i(t) = 0.7*T_i(t-1) + 0.3*P_i(t) across 4 discrete states.

✓ Obj 5 · Adaptive Policy Enforcement
Dynamically gate PRIVILEGED operations (workflow execution, delegation) vs STANDARD operations across trust states.

✓ Obj 6 · Trust-Aware Discovery & Matching
Implement capability discovery and task-aware matching algorithm: MatchScore = 0.4T + 0.3(r/5) + 0.2*sigma + 0.1*(1-lambda).

Rigorous Scope Boundaries & Academic Honesty

IMPLEMENTATION BOUNDARY DECLARATIONS

In accordance with scientific integrity, TrustChain-AI explicitly defines its engineering boundaries:

- **1. Local Relational Storage (Non-Blockchain):**
  The Verifiable Data Registry conforms to W3C VDR specifications via SQLite WAL mode for sub-5ms latency and zero gas fees, rather than a public blockchain.
- **2. Deterministic Agent Simulation:**
  Agent execution is evaluated using LocalAgentConnector for deterministic reproducibility, avoiding external LLM billing costs and stochastic jitter.
- **3. Sequential Pipeline Orchestration:**
  Workflows execute sequentially through ordered nodes. DAG parallel fork-join execution is modeled in schema but reserved for future work.
- **4. Closed-World Evaluation Scale:**
  Tested and validated with 8-10 registered agents; million-agent scale requires distributed DHT indexing.

Five-Tier Security Architecture & Request-Response Dataflow

Layered Architectural Decomposition

Tier 1 · Presentation Tier (Port 5173)

React 19 + Vite frontend styled with Tailwind CSS SOC theme. Provides Dashboard, Agent Explorer, Discovery, Matching, and Admin views.

Tier 2 · API Gateway (Port 8050)

FastAPI ASGI application enforcing CORS middleware, JWT authentication, Argon2id passwords, and security headers (nosniff, DENY).

Tier 3 · Trust Intelligence Core

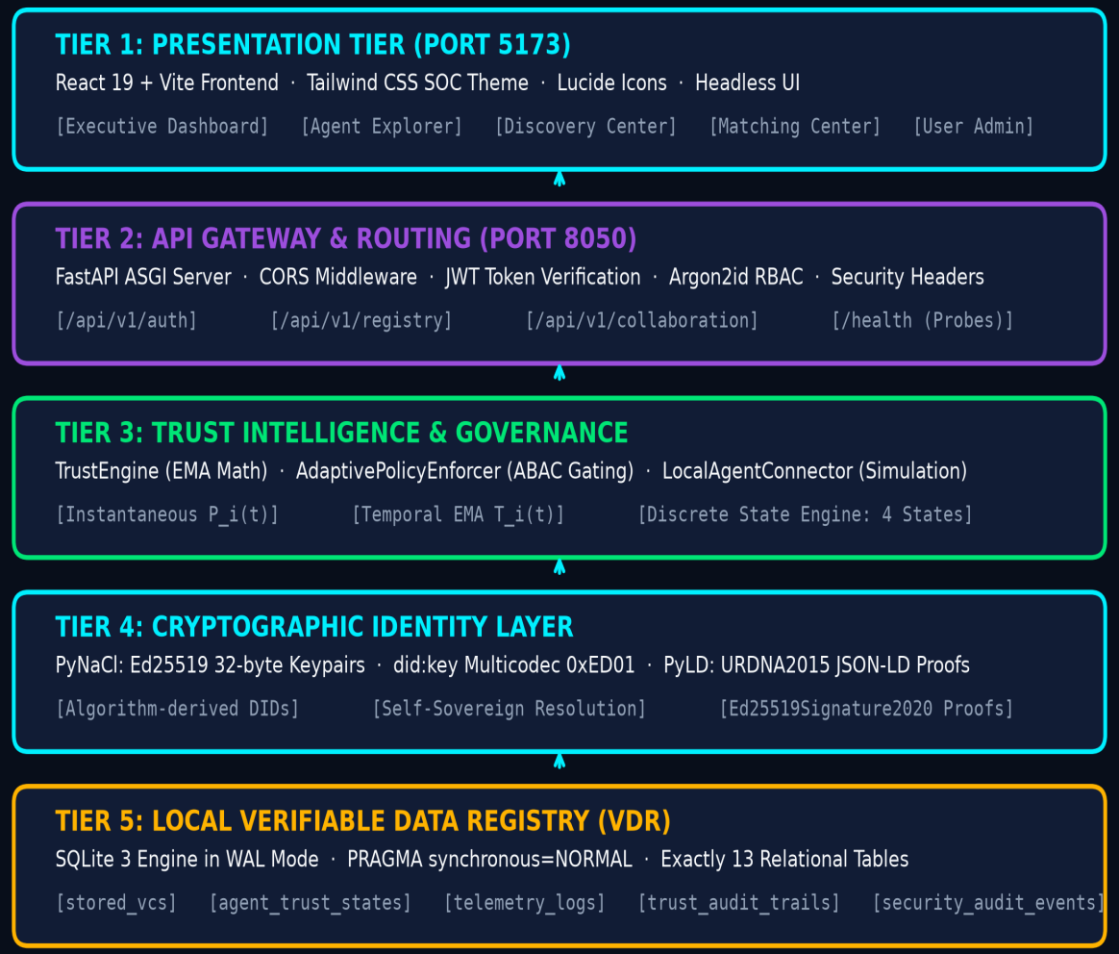
In-process TrustEngine (EMA scoring), AdaptivePolicyEnforcer (RBAC/ABAC gating), and LocalAgentConnector simulation harness.

Tier 4 · Cryptographic Identity

PyNaCl for 32-byte Ed25519 keypairs and did:key generation. PyLD for URDNA2015 JSON-LD credential canonicalization.

Tier 5 · Relational VDR Storage

SQLite in WAL mode (PRAGMA synchronous=NORMAL) across 13 relational tables with sub-5ms read/write latency.



Cryptographic Agent Identity & Verifiable Credentials Lifecycle

W3C did:key Decentralized Architecture

1. Algorithmic Identity Construction

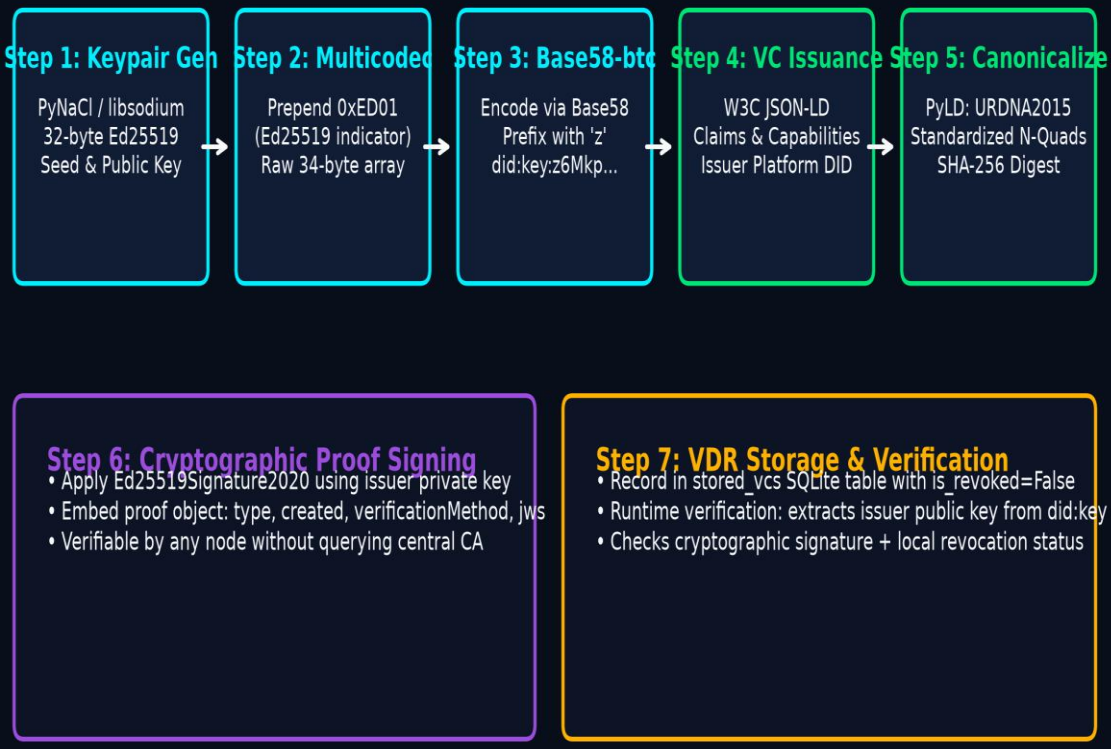
- Raw Keypair: Generates 32-byte Ed25519 seed & public key via libsodium (PyNaCl library).
- Multicodec Prefix: Prepends 2-byte varint 0xED01 indicating Ed25519 public key.
- Base58 Encoding: Encodes prefixed bytes with Bitcoin base58 alphabet, prefixed with 'z'.
- Formal Format: did:key:z6MkpTHR8VNsBxYAAWHut2Geadd9jSwuBV8xRo...

2. Self-Sovereign Decentralization

- No Central Authority: Identifiers are generated algorithmically without requiring DNS registration, central certificate authorities, or registration fees.
- Deterministic Resolution: Any node in the network resolves public key directly from the DID string itself without querying an external registry.

3. Identity vs. Trust Boundary Axiom

Possessing a valid cryptographic did:key proves private key ownership; it provides ZERO guarantee of runtime correctness, safety, or compliance.



Behavioral Telemetry Processing & Temporal EMA Trust Formulation

Mathematical Formulations & Sensor Weights

1. INSTANTANEOUS PERFORMANCE EQUATION

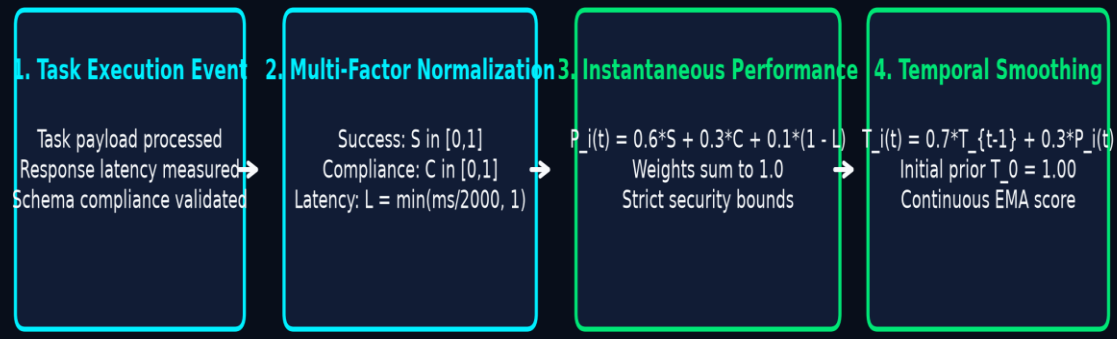
$$P_i(t) = 0.6 \cdot S_i(t) + 0.3 \cdot C_i(t) + 0.1 \cdot (1 - L_i(t))$$

- $S_i(t) \in [0.0, 1.0]$: Functional correctness and task success rate.
- $C_i(t) \in [0.0, 1.0]$: Policy compliance (schema validity, zero prompt leaks).
- $L_i(t) = \min(\text{latency_ms} / 2000, 1.0)$: Latency penalty against 2000ms SLA ceiling.
- Weighting Rationale: Correctness (0.6) and Safety (0.3) are critical enterprise invariants; Latency (0.1) provides marginal optimization.

2. TEMPORAL EMA TRUST EQUATION

$$T_i(t) = 0.7 \cdot T_i(t-1) + 0.3 \cdot P_i(t) \quad [\text{Initial Prior } T_0 = 1.00]$$

- $\alpha = 0.7$: Historical memory retention parameter.
- $\beta = 0.3$: Instantaneous performance responsiveness parameter.
- Prevents single-event volatility while guaranteeing rapid quarantine in 2 consecutive complete failures.



5. Discrete State Classification & Policy Enforcement Action

- T >= 0.80 -> TRUSTED: Full standard & privileged operation authorization granted.
- 0.50 <= T < 0.80 -> SUSPICIOUS: Standard tasks permitted; privileged workflows blocked (HTTP 403).
- T < 0.50 -> QUARANTINED: All operational execution halted; isolated to supervised canary tasks.

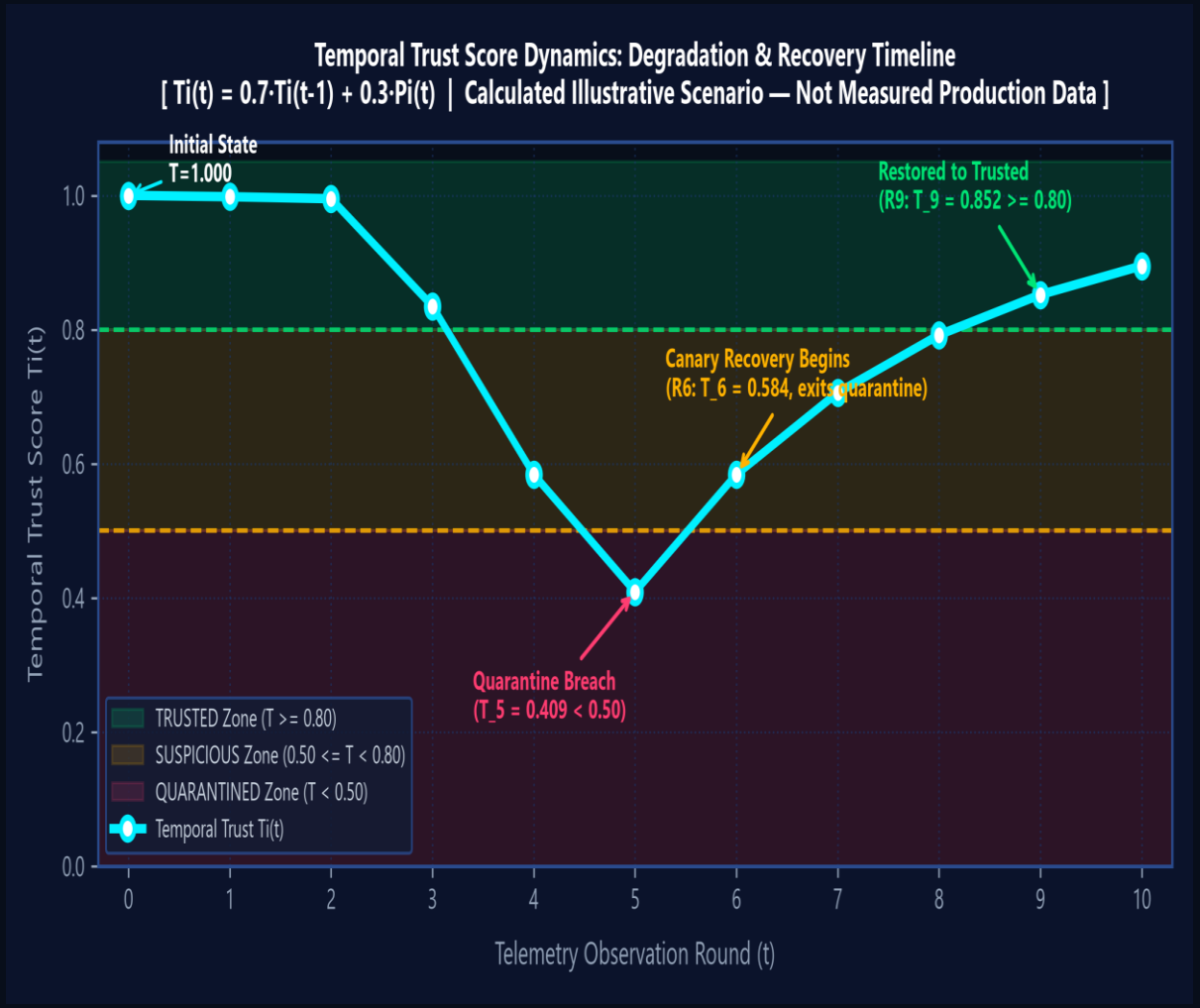
Mathematical Trajectory Dynamics: Degradation & Recovery

10-Round Illustrative Scenario Calculations (T_0 = 1.000)

R0: Baseline Prior: T_0 = 1.000 [State: TRUSTED]
R1: Flawless Task (P=0.994): T_1 = 0.7(1.000) + 0.3(0.994) = 0.998 [State: TRUSTED]
R2: Normal Execution (P=0.988): T_2 = 0.7(0.998) + 0.3(0.988) = 0.995 [State: TRUSTED]
R3: Latency Spike (P=0.460): T_3 = 0.7(0.995) + 0.3(0.460) = 0.835 [State: TRUSTED]
R4: Complete Failure (P=0.000): T_4 = 0.7(0.835) + 0.3(0.000) = 0.584 [State: SUSPICIOUS]
R5: Repeated Failure (P=0.000): T_5 = 0.7(0.584) + 0.3(0.000) = 0.409 [State: QUARANTINED (T<0.50)]
R6: Supervised Canary 1 (P=0.991): T_6 = 0.7(0.409) + 0.3(0.991) = 0.584 [State: SUSPICIOUS]
R7: Supervised Canary 2 (P=0.992): T_7 = 0.7(0.584) + 0.3(0.992) = 0.706 [State: SUSPICIOUS]
R8: Supervised Canary 3 (P=0.993): T_8 = 0.7(0.706) + 0.3(0.993) = 0.792 [State: SUSPICIOUS]
R9: Supervised Canary 4 (P=0.993): T_9 = 0.7(0.792) + 0.3(0.993) = 0.852 [State: TRUSTED (T>=0.80)]
R10: Restored Ops (P=0.994): T_10 = 0.7(0.852) + 0.3(0.994) = 0.895 [State: TRUSTED]

CALCULATED ILLUSTRATIVE SCENARIO — NOT MEASURED PRODUCTION DATA:

- Rapid Quarantine: Isolated in exactly TWO complete failures (T_5 = 0.409 < 0.50).
- Inertial Recovery: Requires FOUR supervised canary rounds (P>0.99) to restore Trusted state (T_9 = 0.852 >= 0.80).



Defense-in-Depth: Human RBAC & Autonomous Agent ABAC

Human RBAC & Defense-in-Depth Security

1. Argon2id Password Hashing

• Memory-Hard Algorithm: Combines Argon2d (GPU/ASIC resistance) and Argon2i (side-channel resistance). OWASP recommended.

2. Five-Strike Account Lockout Policy

• 5 consecutive incorrect passwords trigger an automatic 15-minute lockout (HTTP 423 Locked), mitigating CPU exhaustion brute-forcing.

3. Human Role Hierarchy (Portal Operators)

• ADMIN: Full mutations, user management, manual trust adjustments.
• OPERATOR: Workflow execution, discovery search, task matching.
• VIEWER: Complete read-only observability; mutation controls disabled.
• Frontend badges are UI conveniences; FastAPI backend is SOLE boundary.

HUMAN OPERATOR RBAC (JWT + ARGON2ID)

• ADMIN Role:
Full access: user management, manual trust adjustment, policy configuration, system metrics.

• OPERATOR Role:
Agent execution, task dispatch, discovery filtering, matching review, collaboration monitoring.

• VIEWER Role:
Read-only observability: metrics, audit logs, trust history. Mutation actions disabled in UI.

• 5-Strike Lockout Policy:
5 failed login attempts -> 15-minute lock.
Argon2id memory-hard password protection.

AUTONOMOUS AGENT ABAC (DYNAMIC GATING)

• TRUSTED (T >= 0.80):
ALLOWED: Standard tasks & Privileged workflows.

• SUSPICIOUS (0.50 <= T < 0.80):
ALLOWED: Standard tasks (read, query, map).
BLOCKED (HTTP 403): Privileged workflows.

• QUARANTINED (T < 0.50):
BLOCKED (HTTP 403): All production execution.
ALLOWED: Supervised canary remediation tasks.

• REVOKED (Terminal):
BLOCKED (HTTP 403): Total permanent exclusion.
T locked at 0.0; non-recoverable via telemetry.

Inventory Exploration, Health Probing & Explainable Rationales

Discovery Controls & Explainable Rationales

1. Multi-Parametric Search Controls

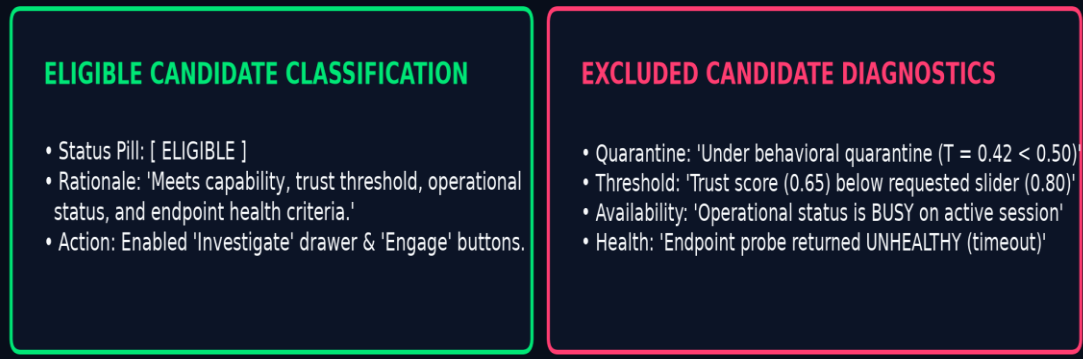
• Capability Tag Selector: Filters by verified credentials (analysis, coding, research, planning, verification).
• Dynamic Min Trust Slider: Real-time slider (0.00 to 1.00 in steps of 0.05).
• Operational Availability Filter: AVAILABLE, BUSY, OFFLINE status.
• Endpoint Health Probe Filter: Live ping confirming HTTP 200 reachable.

2. Explainable Eligibility vs. Exclusion Engine

Unlike opaque directories that silently hide excluded agents, TrustChain-AI explicitly evaluates every registered candidate and displays transparent rationales:
• Quarantine Exclusion: 'Under behavioral quarantine (T = 0.42 < 0.50)'
• Threshold Exclusion: 'Trust score (0.65) below requested slider (0.80)'
• Availability Exclusion: 'Status is BUSY on active collaboration session'

Direct Action Integration: Direct one-click 'Investigate' drawer and 'Engage in Collaboration' buttons.





Task-Driven Matchmaker & Explainable Multi-Factor Ranking

Multi-Factor Matching Algorithm Formulation

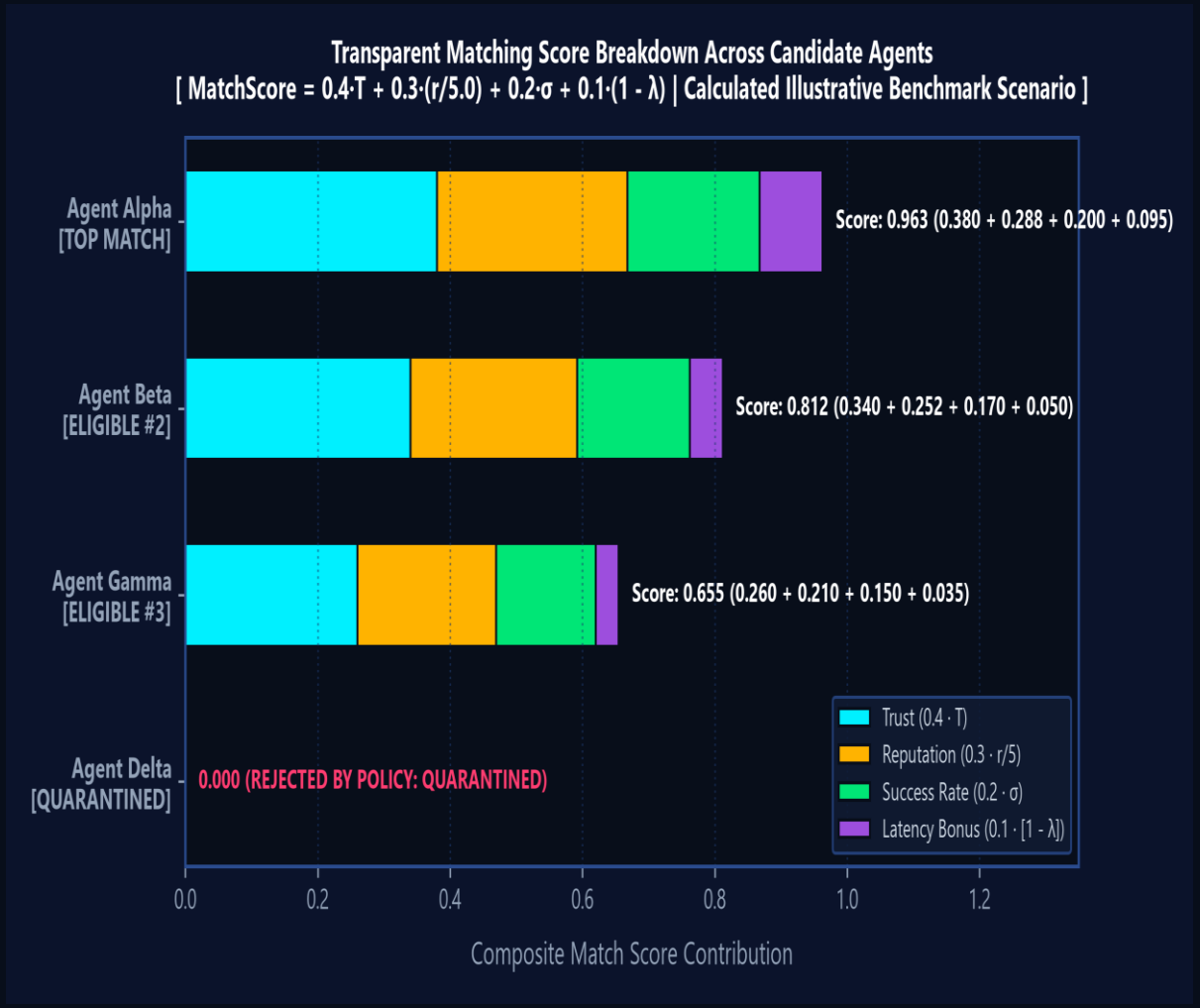
COMPOSITE MATCHSCORE EQUATION

$$\text{MatchScore} = 0.4 \cdot T + 0.3 \cdot (r/5.0) + 0.2 \cdot \sigma + 0.1 \cdot (1 - \lambda)$$

- T: Normalized temporal trust score [0.0, 1.0] (40% weight).
- r: Historical peer reputation [1.0, 5.0] normalized by /5.0 (30% weight).
- σ: Historical task success rate (successes / total_tasks) (20% weight).
- λ: Latency penalty min(latency_ms / 2000, 1.0) -> (1-λ) is latency bonus (10%).

HARD ZERO-TRUST DISQUALIFICATION GATES

- Immediate Disqualification: If state is QUARANTINED or REVOKED, MatchScore is forced to 0.0 with 'REJECTED BY POLICY', regardless of high reputation.
- Candidate Benchmark Breakdown (0.4·T + 0.3·r/5 + 0.2·σ + 0.1·[1-λ]):
  1. Agent Alpha: 0.963 [TOP CANDIDATE SPOTLIGHT] (0.380 + 0.288 + 0.200 + 0.095)
  2. Agent Beta: 0.812 [ELIGIBLE CANDIDATE] (0.340 + 0.252 + 0.170 + 0.050)
  3. Agent Gamma: 0.655 [ELIGIBLE CANDIDATE] (0.260 + 0.210 + 0.150 + 0.035)
  4. Agent Delta: 0.000 [DISQUALIFIED - UNDER QUARANTINE (T=0.42)]
- Calculated illustrative benchmark matching akta/vdr/registry_crud.py.



Multi-Agent Collaboration Sessions & Sequential Pipeline Orchestration

Bilateral Collaboration Protocol & Lifecycle Management

1. Session Lifecycle State Machine

Collaboration sessions progress through deterministic states:
[PENDING] -> [ACTIVE] -> [COMPLETED] / [TERMINATED]

- PENDING: Session initiated, mutual identity & VC exchange verified.
- ACTIVE: Bilateral task payload execution and sensor telemetry logging.
- COMPLETED: Task successfully delivered, mutual ratings recorded.
- TERMINATED: Forced termination upon trust decay or policy breach.

2. Session Data Model (akta/vdr/models.py)

- session_id: UUIDv4 unique session handle.
- initiator_did & target_did: Public did:key identifiers.
- session_token: Cryptographically signed ephemeral JWT.
- collaboration_events: Granular append-only event log capturing every request, step, payload hash, and sensor signature.

3. Mid-Session Trust Boundary Enforcement

If either agent's trust drops below 0.50 during a multi-turn collaboration, the session is instantly aborted, preventing tainted output propagation.



Local Relational VDR Architecture & Quad-Dimensional Auditability

13-Table Relational Schema & WAL High-Throughput Storage

1. Verifiable Data Registry (VDR) Implementation

TrustChain-AI implements a local relational VDR using SQLite configured with Write-Ahead Logging (WAL mode):
• Sub-5ms Read/Write: High-throughput local operations without gas fees, block confirmation delays, or RPC network failures.
• ACID Compliance: Atomic transaction guarantees for all trust state updates and credential issuance operations.

2. Quad-Dimensional Audit Streams

• stored_vcs: Credential lifecycle, issuer signatures, LD-Proofs.
• trust_audit_trails: Immutable state transitions (old_score, new_score, old_state, new_state, trigger_reason, timestamp).
• telemetry_logs: Granular per-task sensor telemetry records.
• security_audit_events: Gateway security events (logins, lockouts, RBAC violations, policy blocks).

3. Academic Clarification: Relational VDR vs Blockchain

Per W3C DID specifications, a Verifiable Data Registry is a logical interface role. SQLite WAL provides high-performance local compliance; distributed Byzantine consensus represents post-capstone future work.

DOMAIN 1: IDENTITY & VCS

- **stored_vcs**
  vc_id, issuer_did, subject_did, claims, proof, is_revoked
- **agents**
  did, name, capabilities, endpoint_url, created_at

DOMAIN 2: TRUST & METRICS

- **agent_trust_states**
  did, trust_score, state, last_evaluated
- **telemetry_logs**
  log_id, agent_did, success, compliance, latency, ...
- **trust_audit_trails**
  trail_id, did, old_score, new_score, trigger_reason, ts

DOMAIN 3: WORKFLOWS & NODES

- **workflows**
  workflow_id, name, status, created_at
- **workflow_nodes**
  node_id, workflow_id, step_order, agent_did, depends_on

DOMAIN 4: COLLABORATION

- **collaboration_sessions**
  session_id, initiator_did, target_did, status, token
- **collaboration_events**
  event_id, session_id, step_index, payload, signature
- **agent_ratings**
  rating_id, rater_did, rated_did, rating_score (1-5), comments

DOMAIN 5: AUTH & SECURITY

- **users**
  user_id, username, password_hash (Argon2id), role, locked_until
- **refresh_tokens**
  token_id, user_id, token_hash, expires_at, revoked
- **security_audit_events**
  event_id, event_type, severity, details, ip_address, ts

Operational SOC Portal Demonstration & Interface Verification

Enterprise SOC Dashboard & Navigation Architecture

1. Modern Frontend Architecture

• React 19 + Vite: Production bundle built in 19.99s (2,478 modules transformed, zero errors).
• Dark Navy SOC Theme: #080E1A background, #111C35 containers, #00F0FF accents, with seamless light mode toggle.
• Lucide Icons & Headless UI: Accessible, keyboard-navigable operational controls.

2. Five Dedicated Operational Modules

• Executive Dashboard: High-level KPI summary, trust distribution pie/bar charts, active collaboration count, live security alert feeds.
• Agent Explorer (/agents): Grid view of all registered agents with real-time status badges (TRUSTED, SUSPICIOUS, QUARANTINED, REVOKED).
• Discovery Center (/discovery): Capability tag selector, dynamic trust threshold slider (0.0–1.0), and diagnostic eligibility badges.
• Matching Center (/matching): Task requirement input, multi-factor ranking, top candidate spotlight card, and explainable disqualification accordion.
• User Management (/admin/users): Role-based access control (ADMIN, OPERATOR, VIEWER) with customized badges and audit logs.

Interactive Investigation Drawer & Forensic Explainability

AGENT INVESTIGATION FORENSIC DRAWER

Clicking 'INVESTIGATE' in Agent Explorer mounts slide-over AgentInvestigationDrawer.jsx, fetching live state via backend APIs:

• Identity & Uptime: Agent DID, model name, provider, execution endpoint, and live 'Test Health' probe button.
• 4 Core KPI Cards: Live Trust Score T(t), Reputation r/5.0, operational status badge, and endpoint health.
• Workload Metrics: Total tasks, successes, failures, success rate %, and average latency ms.
• Trust Math Causality: Real-time P(t) = 0.6·S + 0.3·C + 0.1·(1-L) score breakdown from recent telemetry events.
• Interactive Sub-Tabs: Toggle between 'Recent Telemetry' (task IDs, status, compliance, latency) and 'Trust Transitions' (audit history).
• Deep-Dive Profile: Direct 'Full View' action link route to /agents/{did} (AgentDetails.jsx) with historical trend charts.

EXPLAINABLE DECISION INTERFACE

Matching Center displays explicit mathematical breakdown chips (0.4·T + 0.3·r/5 + 0.2·σ + 0.1·[1-λ]) alongside hard policy disqualification reasons for rejected candidates.

Empirical Testing, Subsystem Health & Mathematical Verification

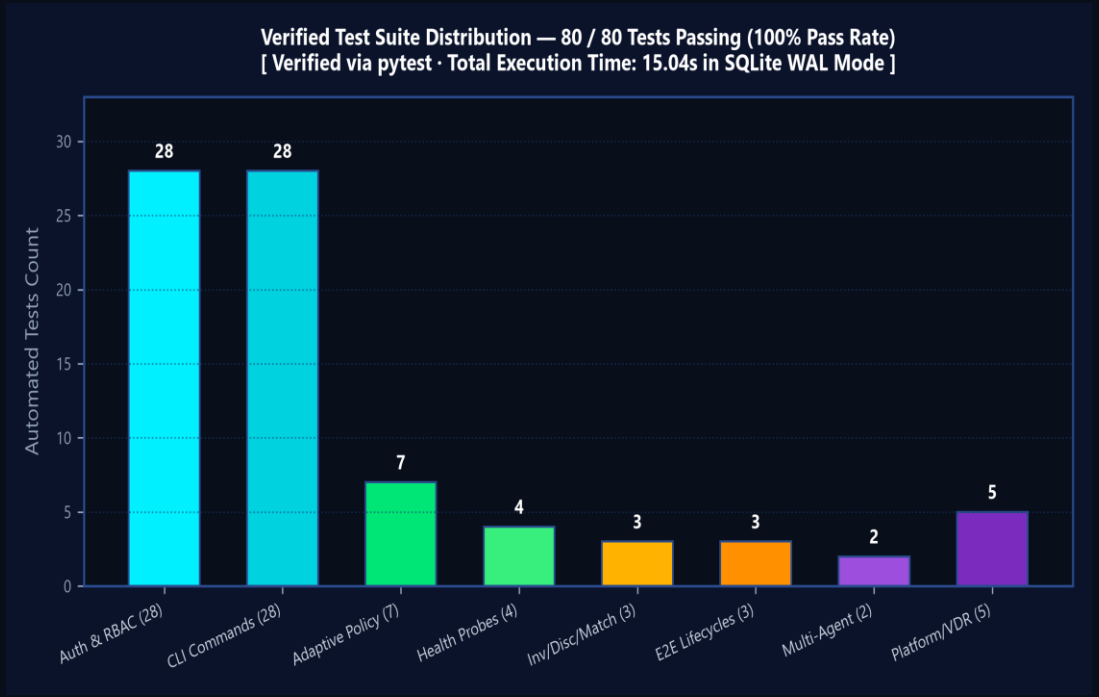
Automated Pytest Suite & 100% Verification Pass Rate

VERIFIED TEST SUITE: 80 PASSED IN 15.04s

Every core framework component is validated by rigorous automated pytest test suites with zero failures and zero regressions:

- ✓ **test_claim.py & test_keys.py (15 tests)**
  Ed25519 key generation, did:key multicodec 0xED01, base58-btc, signature verification
- ✓ **test_vdr.py & test_registry.py (10 tests)**
  W3C VC schema validation, URDNA2015 canonicalization, LD-Proofs, tampering detection
- ✓ **test_adaptive_policy.py (7 tests)**
  RBAC + ABAC enforcement, privileged vs standard action gating, HTTP 403 intercepts
- ✓ **test_authentication.py (28 tests)**
  Argon2id password hashing, JWT refresh rotation, 5-strike lockout, role matrices
- ✓ **test_e2e_lifecycle.py (3 tests)**
  Complete agent onboarding, VC issuance, telemetry stream, and trust progression
- ✓ **test_investigation_discovery_matching.py (3 t)**
  Investigation drawer endpoints, capability discovery filters, exclusion rationales
- ✓ **test_multi_agent.py & test_platform_features.py (10 t)**
  Bilateral session lifecycle, 13-table SQLite VDR operations, matching formula
- ✓ **test_system_health.py (4 tests)**
  Live /health probe, database WAL latency check, trust math core verification

Frontend Build: Vite 8.2.2 production bundle compiled in 14.43s with 0 errors.



Live Subsystem Health Probing (/health)

AUTHORITATIVE HEALTH RESPONSE (GET /health):


```
{
  "status": "ok",
  "components": {
    "database": {"status": "ok", "mode": "WAL", "latency_ms": 0.42},
    "trust_engine": {"status": "ok", "weights": {"w1": 0.6, "w2": 0.3, "w3": 0.1,
"lam": 0.7}},
    "fastapi": {"status": "ok", "port": 8050, "version": "1.0.0"}
  }
}
```

Transparent Limitations, Threats to Validity & Research Roadmap

7 Frank Technical Limitations & Threats to Validity

- **1. Local Relational Storage:** Uses SQLite WAL rather than a distributed Byzantine consensus ledger (Ethereum/Hedera).
- **2. Deterministic Telemetry Emulation:** Evaluated via LocalAgentConnector; wild LLM reasoning variance remains unmeasured.
- **3. Sensor Authenticity Assumption:** Assumes local gateway sensors are uncompromised; internal sensor spoofing is out of scope.
- **4. Static Hyperparameter Weights:** EMA weights (0.7/0.3) and matching weights (0.4/0.3/0.2/0.1) are fixed rather than learned.
- **5. Sequential Pipeline Boundary:** Multi-agent workflows run sequentially; parallel DAG fork-join execution is future work.
- **6. Closed-World Evaluation Scale:** Benchmarked with 8-10 agents; million-agent scale requires distributed DHT indexing.
- **7. False Positive Anomaly Risk:** Transient network hiccups can cause brief latency penalties, temporarily nudging trust into Suspicious.

Strategic Future Research Roadmap & Scalability Horizon

PHASED RESEARCH & EVOLUTION ROADMAP

TrustChain-AI establishes a strong foundation. Four key enhancements define the future research trajectory:

[PHASE 1 · DISTRIBUTED BYZANTINE VDR]
Integrate Hedera Hashgraph or Tendermint consensus to replace local SQLite with decentralized, multi-region Byzantine fault tolerance.

[PHASE 2 · ZERO-KNOWLEDGE TELEMETRY PROOFS]
Deploy zk-SNARKs so proprietary commercial AI agents can mathematically prove task compliance without disclosing proprietary model prompts or training weights.

[PHASE 3 · REINFORCEMENT LEARNING WEIGHT ADAPTATION]
Replace static weighting constants ($\alpha=0.7$, $\beta=0.3$) with context-aware reinforcement learning agents that tune weights based on environmental volatility.

[PHASE 4 · ASYNCHRONOUS DAG WORKFLOW ENGINE]
Upgrade sequential pipeline engine to support full DAG fork-join parallel execution with asynchronous synchronization barriers.

Capstone Defense Conclusion, Core Contributions & Open Defense

Summary of Three Core Research & Engineering Contributions

1. Cryptographic Trust Anchor (W3C did:key & VCs)

Engineered decentralized agent identity using Ed25519 keypairs and multicodec encoding without central CA reliance. Implemented W3C Verifiable Credentials with URDNA2015 canonicalization and Ed25519Signature2020 proofs.

2. Dynamic Behavioral Telemetry & Adaptive Zero-Trust Gating

Replaced static binary authentication with continuous Exponential Moving Average trust scoring ($T = 0.7 \cdot T + 0.3 \cdot P$). Coupled trust scores directly to operation-level adaptive gating (blocking privileged actions in Suspicious, total quarantine below 0.50).

3. Verifiable Traceability & Explainable Matching

Built a 13-table SQLite WAL Verifiable Data Registry with quad-dimensional auditability. Created an explainable multi-factor candidate matching engine ($MatchScore = 0.4T + 0.3(r/5) + 0.2\sigma + 0.1(1-\lambda)$) backed by a modern React 19 SOC portal.

Empirical Validation: 80/80 passing pytest tests in 15.04s and 100% verified production Vite frontend build (14.43s).

Capstone Defense Conclusion & Committee Q&A

PROJECT DEFENSE SUMMARY

TrustChain-AI delivers a complete, verifiable, closed-loop zero-trust architecture for autonomous AI agents. By replacing fragile static authentication with continuous behavioral evaluation, TrustChain-AI guarantees that autonomous agent collaboration remains secure, compliant, and accountable even in the presence of runtime model degradation or adversarial drift.

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DEFENSE COMMITTEE Q&A SESSION

Thank you for your time and evaluation.
We welcome questions from the examination committee.